



Multi-task learning boosted predictions of the remaining useful life of aero-engines under scenarios of working-condition shift

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ABSTRACT

The aero-engine is a typical equipment operating under variable working conditions. Changes in the working conditions of an aero-engine can cause data distribution divergence, making the remaining useful life (RUL) prediction task more challenging. Previous domain adaptation (DA) approaches have the limitation on the prerequisite of data availability in the target domain when handling the domain discrepancy and arranging data alignment. The target working condition is more likely to be unseen, resulting in the unavailability of the corresponding condition monitoring data of this working scenario. This study presents the research topic: the RUL prediction of aero-engines under working-condition shift scenarios in the absence of target domain data. To this end, we propose a multi-task learning-boosted method (MTLTrans) for the cross-domain RUL prediction of aero-engines. The MTLTrans is built upon the Transformer backbone in a hierarchical sharing style with two auxiliary prognostics tasks, i.e., state of health (SOH) assessment and performance degradation (PD) prediction. The trade-off learning of these three tasks facilitates producing reliable RUL prediction results robust against the data shift. Experiments on 12 cross-domain scenarios have shown that the proposed method significantly outperforms state-of-the-art methods, with an improvement of 18.83% of the root mean square error (RMSE).

1. Introduction

The aero-engine is viewed as the heart of an aircraft, and its reliability and safety are of great significance, since the failure of the aero-engine can cause severe economic losses and casualties. Remaining useful life (RUL) prediction is one of the essential tasks in the prognostics and health management (PHM) [1–4]. Aero-engines usually experience complex operating conditions and varieties of fault modes, and thereby, the collected condition monitoring data are represented by different distributions, as shown in Fig. 1. In this case, the marginal distribution of data can vary associated with the variation of the working conditions. Most data-driven methods ignore the divergence of the distribution of the sensor signals from training to the test, i.e., they assume the same distribution of training and test data. Consequently, this can cause significant underperformance of data-driven methods. Indeed, the same distribution does not hold in real applications due to variations and uncertainties in the working conditions of aero-engines, such as uncertainty of operating conditions and dynamic flight jobs. What makes it worse is that there is little historical data available

on the target working condition, which hinders the data alignment of the source domain and target domain under different working conditions. How to tackle this working-condition shift problem with little target domain data remains a major obstacle for the RUL prediction of aero-engines.

In essence, the working-condition shift problem of aero-engines results in the data shift problem for the RUL prediction model. Domain adaptation (DA) is an efficient way of tackling the problem of different data distributions by transferring knowledge learned in the source domain to the target domain. The DA methods generalize the domain-invariant features learned to the target domain using the projection of training and test data in the feature space to the same new subspace. Generally, for the working-condition shift problem in the RUL prediction task, existing DA methods can be classified into four main categories, i.e., instances-based, parameters-based, adversarial-based and statistics-matching-based. Instances-based methods attempt to minimize the data distribution discrepancy between the source and

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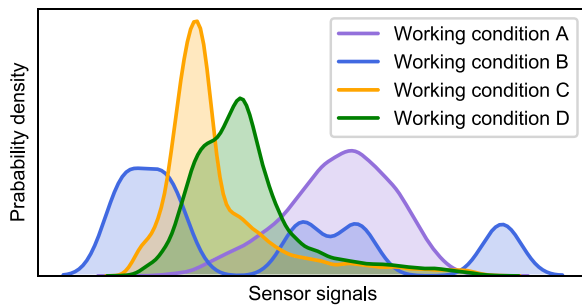


Fig. 1. Probability density distribution of sensor signals of aero-engines under different working conditions.

Source: The data comes from the C-MAPSS [5].

target domains by re-weighting certain source instances and viewing them as a supplement of target data. Similarly, parameters-based methods reuse and fine-tune parameters pre-trained in the source domain. Nevertheless, these two kinds of DA methods demand the accessibility of labeled data from the target domain. This is not feasible in real-world scenarios as labeling data is difficult, expensive, time-consuming, error-prone, and requires substantial human labor and expertise. In the case of scarce labeled target data, instances-based and parameters-based methods would suffer the risk of over-fitting.

The adversarial-based and statistics-matching-based methods eliminate the demand for labeled target data via unsupervised learning. The adversarial-based methods exploit the characteristics of the generative adversarial network (GAN) to achieve adversarial training. For example, Ragab et al. [6] and Zhuang et al. [7] proposed an adversarial-based method for the cross-domain RUL prediction of aero-engines, in which the adversarial learning and contrastive learning are integrated to capture domain-invariant features and preserve the target-specific mutual information. Considering the different degradation characteristics of the sensors on different aero-engines, Chen et al. [8] proposed a multi-level feature learning strategy via adversarial ensemble learning. On the contrary, compared with the min-max optimization problems in adversarial-based methods, statistics-matching-based methods focus on the minimization optimization of the distribution discrepancy. For example, Li et al. [9] defined the representation subspace distance and designed a shrinkage attention module to eliminate irrelevant information and increase the focus on critical features. Jin et al. [10] introduced a multi-layer multi-kernel maximum mean discrepancy (MMD) to reduce the distribution discrepancy of different domains via a domain adaptive residual network. Similarly, improved MMD has been proposed in [11–13]. Furthermore, several researchers have tried to combine adversarial-based and statistics-matching-based methods. For instance, Shi et al. [14] introduced Wasserstein distance as the metric of domain distribution difference to enhance the robustness of adversarial training and avoid the problems of gradient disappearance. A modified MMD metric and adversarial mechanism are integrated to reduce the domain discrepancy based on deep residual long short-term memory (LSTM) [15]. However, although these methods can obtain knowledge transfer and yield promising results, they cannot be competent when the unlabeled data of the target domain is scarce or even unavailable. The distribution discrepancy metrics can be meaningless because we cannot characterize the actual underlying distribution of the target domain with little target data. The data availability of the target domain limits the prediction performance of DA-based methods.

As training a reliable DA model in the absence of target domain data is hard to accomplish, alternatives that do not depend on the target domain data have gradually been discussed. These methods directly improve the generalization capability and robustness of the RUL prediction model against scenarios where new operating conditions and new faults are present in the unseen target data. Compared with

statistics-matching-based DA methods, they do not directly explore the minimum optimization of the distribution discrepancy but focus on the robustness training. For example, Xiao et al. [16] intentionally injected additive white Gaussian noise into the degradation features fed to a bidirectional LSTM (BiLSTM) network to enhance the robustness of the RUL prediction under different working conditions. Besides, Ding et al. [17] deploy an adversarial generator for out-domain augmentation (AOA) that maximizes the difference and diversity of the generated pseudo-domains to mimic possible unseen targets. The AOA framework achieved domain generalization (DG) over the target domains. Likewise, [18] proposed an adversarial domain generalization framework with regularization learning to mine latent domains.

On the other hand, multi-task learning (MTL) also has the potential to extract generally shared representations and obtain robust performance [19]. In MTL, the associated tasks share complementary information or act as a regularizer for one another, making it possible to explore the latent feature representations. For the noise-rich sensor signals of aero-engines, the multiple tasks in the MTL structure can handle different noise patterns by leveraging the task-specified information. In Ross's study, through the shared representation, the simultaneous learning of multiple related tasks can neutralize some of the noise and generate robust results compared to the independent learning for each task [20]. Wang et al. [21] designed a feature extraction task to integrate with the RUL prediction task by a convolutional neural network (CNN) and a self-attention encoder. Similarly, Kim et al. [22] constructed an MTL model involving health condition identification and RUL prediction via deep CNNs. Zhang et al. [23] combined the health status assessment and RUL prediction based on bidirectional gated recurrent unit (BiGRU) and multi-gate mixture-of-experts (MMoE). In [24], the first predicting time (FPT) determination and RUL prediction are combined to generate entity-invariant features for the RUL prediction of rotating machines. Nevertheless, auxiliary tasks among these works are usually predecessor tasks of RUL prediction, i.e., features extraction, health condition identification and FPT determination, making the two tasks highly dependent.

In this work for the RUL prediction of aero-engines under working-condition shift scenarios, a multi-task learning (MTL) architecture is constructed to improve the robustness and generalizability in the absence of target domain data. A Transformer-based encoder-decoder structure is employed as the backbone of the MTL-boosted RUL prediction model (MTLTrans). Two auxiliary prognostic tasks, i.e., state of health (SOH) assessment and performance degradation (PD) prediction, along with the main task of RUL prediction, are successively accomplished in the MTLTrans in a hierarchical sharing manner. In particular, a feature distiller and three diverse task-specified classifiers or predictors are introduced in the MTLTrans to produce generalized and shift-invariant features against the working-condition shift. The MTLTrans enhances the robustness, generalization and applicability of the RUL prediction of aero-engines under working-condition shift scenarios without the demand for data availability.

The key contributions of this work are three-fold:

- We introduce two auxiliary tasks, i.e., state-of-health (SOH) assessment and performance degradation (PD) prediction tasks in a multi-task learning (MTL) model (MTLTrans). This is the first primary exploration of the MTL techniques tackling the working-conditional shift problem for the RUL prediction of aero-engines.
- We accomplish feasible cross-domain RUL prediction against data shift in the absence of target domain data, which meets the demand of practical applications where such data are typically unavailable.
- We construct a hierarchical MTL model based on Transformer architecture to learn the spatial-temporal dependencies and generate the shared feature representations.

The remainder of the paper is organized as follows. Section 2 presents the problem formulation and Section 3 details the framework of the proposed method. Experiments are conducted in Section 4 and we conclude this paper in Section 5.

2. Problem formulation

We define a feature domain $\mathcal{D} = \{\mathcal{X}, P(X)\}$, where $\mathcal{X}$ is the feature space, $X \in \mathcal{X}$ are sensory data, and $P(X)$ is the marginal probability distribution of the sensory data X in the feature space $\mathcal{X}$. The task $\mathcal{T}$ on X defines the label space $\mathcal{Y}$ with the conditional probability distribution $P(Y|X)$, where $Y \in \mathcal{Y}$.

Specifically, in the source domain, we have the labeled training set $\mathcal{D}_S = \{\mathcal{X}_S, P(X_S)\}$ of sensory data from aero-engines in known working conditions. There are available N_S labeled run-to-failure (RtF) trajectories and corresponding RULs of aero-engines $\{X_i^S, y_i^S\}_{i=1}^{N_S}$, where $X_i^S \in \mathbb{R}^{M \times T_i^{rtf}}$ and $y_i^S \in \mathbb{R}^{T_i^{rtf}}$, are the high-dimensional condition monitoring data of sensor signals and their associated RULs of aero-engines, respectively. N_S is the total number of aero-engines in the considered working condition. M is the number of sensors and T_i^{rtf} is the life span of the i th aero-engine.

On the other hand, in the target domain, without loss of generality, the testing engines are unlabeled with their RULs, $\mathcal{D}_T = \{\mathcal{X}_T, P(X_T)\}$, $\{X_i^T\}_{i=1}^{N_T}$ and $X_i^T \in \mathbb{R}^{M \times T_i^{ud}}$ are Up-to-Date(UtD) trajectories before failure. T_i^{ud} is the length of the UtD trajectory of the i th engine. Note that, in practice, $N_T \ll N_S$ and $T_i^{ud} \ll T_i^{rtf}$, $\forall X_i^S \in \mathcal{X}_S$ and $X_i^T \in \mathcal{X}_T$, which makes it impossible to employ the target data to improve the model. N_T is the total number of aero-engines in the unknown, different working conditions.

In our problem, $\mathcal{D}_S$ and $\mathcal{D}_T$ differ by multivariate time-series data of aero-engines developing different operating conditions or fault patterns. Particularly, we have labeled data from aero-engines with a known specific working condition $P(X_S)$, and we aim to improve the RUL prediction performance for engines operating under different working conditions with unlabeled condition monitoring data $P(X_T)$. The different working conditions lead to a significant discrepancy between the source and target datasets: $P(X_S) \neq P(X_T)$.

Through exploration of the training data $\{X_i^S, y_i^S\}_{i=1}^{N_S}$, this study aims at finding a latent feature space $\mathcal{X}'$ in which $P(X_S)$ and $P(X_T)$ are similar. On this basis, a robust RUL prediction model $\hat{y}_i^S = F(w(x_i^S))$ is constructed, where $\hat{y}_i^S$ is the predicted RUL value. The time window strategy is adopted to catch the temporal relationship in the time series. $w(x_i^S) = x_{i-ws+1:t}^S = (x_{i-ws+1}^S, \dots, x_t^S)$ is the time window of the multi-sensor signals at time t , and ws denotes the time window size. In the testing scenarios, $\{X_i^T\}_{i=1}^{N_T}$, the established model is expected to achieve reliable prognostic performance under the working-condition shift problem by $\hat{y}_i^T = F(w(x_i^T))$. y_i^T and $\hat{y}_i^T$ denote the ground truth and the predicted RULs of target domain, respectively. For the sake of clarity, we omit the superscript (S and T for source domain and target domain) and the subscript (t for time), and thus, $w(x_t)$, $\hat{y}$ and y are used from now on. The main symbols are summarized in Table 1.

3. Proposed method

In this paper, SOH assessment and PD prediction tasks are chosen to be the auxiliary tasks. They are essentially a classification problem and a regression problem of multi-step prediction, enriching task types in the MTL models. Besides, the results of the three tasks of MTLTrans, i.e., SOH assessment, RUL prediction and PD prediction tasks, can provide the manager with comprehensive health information about the aero-engine from different perspectives. It can contribute significantly to a reliable and cost-effective (predictive maintenance) PdM decision, allowing flexible operation and maintenance (O&M).

A hierarchical MTL structure is constructed based on an encoder-decoder skeleton of the Transformer. SOH assessment, RUL prediction and PD prediction tasks are accomplished successively. Fig. 2 presents the overview of the proposed MTL-boosted RUL prediction method, and there are four main steps in MTLTrans:

Step 1: Shared Transformer-based Encoder. The RtF trajectories are split into time windows; then, after positional embedding,

they are fed into a shared Transformer-based encoder to generate a concatenated feature map.

Step 2: Task A State-of-health (SOH) Assessment. Taking the concatenated feature map as input, the task-specified layers of the SOH assessment task generate the SOH assessment results.

Step 3: Task B RUL Prediction. In this step, the concatenated feature map of step 1 has been further distilled by the distiller, and the RUL prediction results are yielded through its task-specified layers.

Step 4: Task C Performance Degradation (PD) Prediction. The PD prediction task is accomplished in this step through a Transformer-based decoder and its task-specified layers.

3.1. Framework

In MTLTrans, the shared blocks (i.e., the Transformer-based encoder and the feature distiller) extract the general features of corresponding domains. In contrast, the specified blocks (i.e., task-specified layers and the Transformer-based decoder) focus on performing individual tasks.

For the type of neural network for the backbone of the MTL model for the RUL prediction of aero-engines, the recurrent structure, such as recurrent neural networks (RNNs), LSTM, GRU, BiLSTM and BiGRU, show distinct advantages in capturing the high-dimensional and temporal dependence characteristics among the condition monitoring data [25]. Recently, to empower these models to handle the local dependence and enhance the deep feature learning capability, the attention mechanism has been introduced into the networks [26,27]. Nevertheless, Transformer [28] and its variants, such as Informer [29], have attempted to use the attention mechanism to construct the prognostics model directly, avoiding the sequential computing and associated high computational time expenses caused by the recurrent networks. Therefore, in our study, we employ the Transformer as the backbone of the MTL structure.

3.1.1. Step 1: Shared Transformer-based encoder

Transformer is a classical sequence-to-sequence architecture that consists of an encoder and a decoder, as proposed in natural language processing (NLP). Positional encoding is a classic trick in Transformer to offer the front and back position relationships in the input time series:

$$\begin{aligned} p_t^{(2s)} &= \sin(t/10000^{2s/d}), \\ p_t^{(2s+1)} &= \cos(t/10000^{2s/d}), \end{aligned} \quad (1)$$

where $s \in \{1, \dots, \lfloor d/2 \rfloor\}$, p_t denotes the position encoding w.r.t. time step t and d is the feature dimension after the input representation. In this way, for any given fixed time step l , p_{l+l} has a linear relationship with p_l . As a result, the local context is preserved, and the model can learn to reach the relative position effortlessly.

After positional embedding, the encoder maps the input sequence into a higher dimensional vector. Unlike the sequential nature of RNNs, Transformer builds long-term dependencies via a dot-product-based attention mechanism. The *ProbSparse* self-attention mechanism proposed in Informer efficiently replaces the canonical self-attention, which allows each key to only attend to u dominant queries:

$$\mathcal{A}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Softmax}\left(\frac{\overline{\mathbf{Q}}\mathbf{K}^\top}{\sqrt{d}}\right)\mathbf{V}, \quad (2)$$

where u is the number of active queries, $\overline{\mathbf{Q}}$ is a sparse matrix of the same size of $\mathbf{q}$ and it only contains the **Top- u** queries under the sparse measurement $M(\mathbf{q}, \mathbf{K})$. Sampling factor c controls the time complexity and memory usage on dependency alignments.

Following each *ProbSparse* self-attention layer is a self-attention distilling layer made of a Conv1d and a MaxPooling. This strategy efficiently preserves the dominant features and speeds up calculations. This solves the problem that the encoder's feature map has redundant

Table 1
Main symbols and their description.

Symbols	Description
D_S/D_T	the source/target domain
$P(X_S)/P(X_T)$	the marginal distribution of source/target data (condition monitoring sensor signals)
$\mathcal{X}_S/\mathcal{X}_T$	the feature space of source/target data (condition monitoring sensor signals)
N_S/N_T	the total number of the aero-engine units in the source/target domain
i/t	the index of aero-engine units/cycle time
M	the total number of sensors
X_i^S/X_i^T	the time series for aero-engine unit i in the source/target domain
T_i^{rf}/T_i^{ud}	the length of the trajectories of the i th aero-engine in the source/target domain
$\mathcal{X}'$	a latent feature space
x_i^S/x_i^T	the sensory time series at time t in the source/target domain
F	the RUL prediction function
w_S	the time window size
$w(x_t)$ or $w(x)$	the time window of the sensory time series at time t
$y_t^S/\hat{y}_t^S$	the ground truth/predicted RUL at time t in the source domain
$y_t^T/\hat{y}_t^T$	the ground truth/predicted RUL at time t in the target domain
$y/\hat{y}$	the ground truth/predicted RUL (at time t)
$\mathcal{Z}$	the shared encoder in MTLTrans
$\theta_{\mathcal{Z}}/\zeta$	the parameters/output of $\mathcal{Z}$
$o/\hat{o}$	the ground truth/assessed health state label
t_c/t_e	the time of the incipient failure and the beginning time of the severe failure
$\mathcal{O}$	the SOH classifier
θ_o	the parameters in $\mathcal{O}$
$\mathcal{L}_o$	the classification error of the SOH assessment task
$\mathcal{K}$	the feature distiller
$\theta_{\mathcal{K}}$	the parameters in $\mathcal{K}$
κ	the distilled features
h	the health index of aero-engines
$\mathcal{Y}$	the RUL predictor
θ_y	the parameters in $\mathcal{Y}$
$\mathcal{L}_y$	the prediction error of the RUL prediction task
$\mathbf{x}'$	the masked sensory data
l_m	the length of the placeholder in PD prediction
l_p	the PD prediction horizon
$\mathcal{G}$	the Transformer-based decoder
$\theta_{\mathcal{G}}/g$	the parameters/output of $\mathcal{G}$
$\mathcal{H}$	the performance degradation predictor
θ_h	the parameters of $\mathcal{H}$
$\mathcal{L}_h$	the prediction error of the PD performance task
$\mathcal{L}$	the overall loss of MTLTrans
$\alpha_o/\alpha_y/\alpha_h$	the trade-off parameters of the three tasks
η	the learning rate
y_{max}	the maximum RUL

combinations of value $\mathbf{V}$. The distilling operation from the j th layer to the $j+1$ th layer is:

$$X_{j+1}^T = \text{MaxPool}(\text{ELU}(\text{Conv1d}([X_j^T]_{AB}))), \quad (3)$$

where $[\cdot]_{AB}$ is the attention block and it contains the multi-head *Prob-Sparse* self-attention and the essential operations. $\text{Conv1d}(\cdot)$ performs a 1-D convolutional filter (kernel width=3) on the time dimension with the $\text{ELU}(\cdot)$ activation function. To enhance the robustness of the distilling operation, replicas of the main stack with halving inputs are built and progressively decrease the number of self-attention distilling layers by dropping one layer at a time, like a pyramid, such that their output dimension is aligned. By so doing, all the stacks' outputs are concatenated and have the final hidden representation of the encoder.

A focused self-attention feature map is generated after several stacked self-attention layers and distilling layers in the encoder. For simplification, we use $\mathcal{Z}$ to denote the shared Transformer-based encoder, and ζ and $\theta_{\mathcal{Z}}$ are the corresponding feature representations obtained by $\mathcal{Z}$ and the parameters in the encoder $\mathcal{Z}$. Then, it could be formulated as:

$$\zeta = \mathcal{Z}(w(x); \theta_{\mathcal{Z}}). \quad (4)$$

3.1.2. Step 2: Task A state-of-health (SOH) assessment

The SOH assessment method could monitor the health status of the aero-engine and provide early warning of the incipient failure and server failure, which facilitates the PdM decisions with the health

perception of the aero-engine at present [30]. Compared with the RUL prediction, the SOH assessment is a more general evaluation, as the SOH assessment only focuses on the current health status of the aero-engine [31]. Whereas the RUL prediction should take more consideration to the health status in the long term and in which the working conditions are unknown.

We assume that the aero-engine experiences three stages (Fig. 3): health, initial degradation and severe degradation. In the health stage, the aero-engine works normally, and the sensory data goes up and down slightly without any obvious uptrend or downtrend. Next is the initial degradation stage and it begins at the incipient failure time. In this stage, an uptrend or downtrend could be observed. Finally, it is the severe degradation stage where the aero-engine is severely degraded with a significant uptrend or downtrend of the sensor signals. Eq. (5) formulates these three stages, in which t_c and t_e denote the time of the incipient failure and the beginning time of the severe failure, respectively. The boundary between the health stage and the initial degradation stage is the incipient failure time t_c . In the initial degradation stage, an uptrend or downtrend could be observed. The boundary (t_e) between the initial degradation stage and the severe degradation stage can be defined as the time when the uptrend or downtrend becomes sharp. Usually, the time when $h_i = 0.5$ is the boundary of these two stages, where h_i is the health index (HI) of the

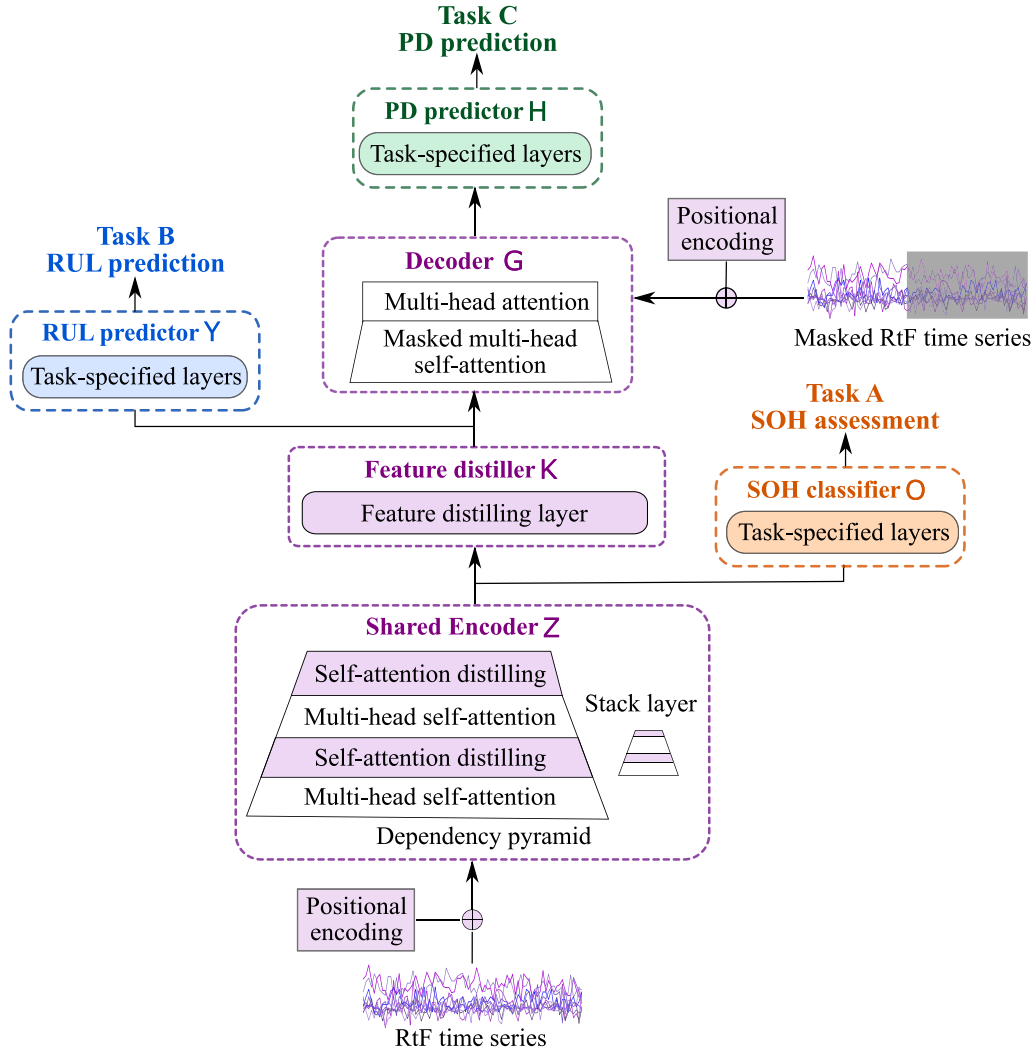


Fig. 2. The framework of the proposed method MTLTrans. A hierarchical MTL structure is constructed based on an encoder–decoder skeleton of Transformer. SOH assessment, RUL prediction and PD prediction tasks are accomplished successively.

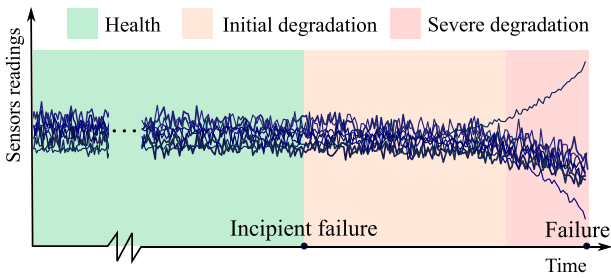


Fig. 3. The three health stages of the aero-engine: health, initial degradation and severe degradation.

Source: The RtF data comes from the C-MAPSS [5].

aero-engine.

$$o_t = \begin{cases} 0, & t \leq t_e, \\ 1, & t_c > t > t_e, \\ 2, & t \geq t_c. \end{cases} \quad (5)$$

In essence, SOH assessment is a classification problem. Thus, we consider the task-specified layers as a classifier. Let $\mathcal{O}$ and θ_o denote the classifier and its parameters, respectively, and then, we can get the

SOH assessment results $\hat{o}$ as:

$$\hat{o} = \mathcal{O}(\mathcal{Z}(w(\mathbf{x}); \theta_\zeta); \theta_o). \quad (6)$$

The classifier $\mathcal{O}$ aims to precisely assess the SOH of the aero-engine through the concatenated feature map ζ drawn from the input sensory time series. Using $\mathcal{L}_o$ to denote the classification error of the SOH assessment task ($\mathcal{L}_o(\hat{o}, o)$), the parameters are updated to achieve the following:

$$\begin{aligned} \hat{\theta}_o &= \arg \min_{\theta_o} \mathcal{L}_o(\mathcal{O}(\mathcal{Z}(w(\mathbf{x}); \hat{\theta}_\zeta); \theta_o), o), \\ \hat{\theta}_\zeta &= \arg \min_{\theta_\zeta} \mathcal{L}_o(\mathcal{O}(\mathcal{Z}(w(\mathbf{x}); \theta_\zeta); \hat{\theta}_o), o). \end{aligned} \quad (7)$$

3.1.3. Step 3: Task B RUL prediction

The output of the shared Transformer-based encoder is a concatenated feature map full of the degradation features in the RtF time series. The feature distiller $\mathcal{K}$ aims to reduce the dimension of the concatenated high-dimensional feature map κ , which reduces the number of parameters and increases the shift-invariant property. The feature distiller represents the main degradation characteristics and prevents redundant information, improving robustness against the working-condition shift problem. $\mathcal{K}$ in Eq. (8) describes this process, where θ_k is the parameters in this feature distilling step:

$$\kappa = \mathcal{K}(\mathcal{Z}(w(\mathbf{x}); \theta_\zeta); \theta_k). \quad (8)$$

In our work, an end-to-end RUL prediction is conducted. In essence, the RUL prediction task (Task B) is a regression problem, and we denote it as:

$$\hat{y} = \mathcal{Y}(\mathcal{K}(\mathcal{Z}(w(\mathbf{x}); \theta_{\zeta}); \theta_{\kappa}); \theta_y), \quad (9)$$

where $\hat{y}$ is the predicted RUL, $\mathcal{Y}$ and θ_y are the RUL predictor and its parameters, respectively.

For simplicity, the RUL predictor $\mathcal{Y}$ receives the features κ from the distiller $\mathcal{K}$ and infers the RUL prediction $\hat{y}$. Using $\mathcal{L}_y$ to denote the prediction error of the RUL prediction task ($\mathcal{L}_y(\hat{y}, y)$), the parameters of the predictor $\mathcal{Y}$ are optimized to achieve the following:

$$\begin{aligned} \hat{\theta}_y &= \arg \min_{\theta_y} \mathcal{L}_y(\mathcal{Y}(\mathcal{K}(\mathcal{Z}(w(\mathbf{x}); \hat{\theta}_{\zeta}); \hat{\theta}_{\kappa}); \theta_y), y), \\ \hat{\theta}_{\kappa} &= \arg \min_{\theta_{\kappa}} \mathcal{L}_y(\mathcal{Y}(\mathcal{K}(\mathcal{Z}(w(\mathbf{x}); \hat{\theta}_{\zeta}); \theta_{\kappa}); \hat{\theta}_y), y), \\ \hat{\theta}_{\zeta} &= \arg \min_{\theta_{\zeta}} \mathcal{L}_y(\mathcal{Y}(\mathcal{K}(\mathcal{Z}(w(\mathbf{x}); \theta_{\zeta}); \hat{\theta}_{\kappa}); \hat{\theta}_y), y). \end{aligned} \quad (10)$$

3.1.4. Step 4: Task C performance degradation (PD) prediction

We adopt the decoder structure in Informer [29], composed of a stack of two identical multi-head attention layers. We feed the decoder with the masked sensory data $\mathbf{x}'$ (Eq. (11)) and the distilled features κ .

$$\mathbf{x}' = \text{Concat}(x_{t-lm+1:t}, x_{t+1:t+lp}^{\text{mask}}). \quad (11)$$

The sensory time series $x_{t+1:t+lp}^{\text{mask}}$ are masked/padded with zero-matrix, whereas the positional embedding is reserved to be a placeholder for the target HI inference. The lm long sequence before t is sampled as an earlier slice before the target output. lp denotes the PD prediction horizon.

The masked RtF time series $\mathbf{x}'$ is also fed into the decoder along with the distilled features to enrich the performance degradation information. Masked multi-head attention is applied in the ProbSparse self-attention computing by setting masked dot products to $-\infty$. It prevents each position from attending to coming positions, which avoids the auto-regressive phenomenon. Thereby, we represent the decoder process by Eq. (12), where $\mathcal{G}$, θ_g and g denote the Transformer-based decoder, the parameters and the output, respectively.

$$g = \mathcal{G}(\mathcal{K}(\mathcal{Z}(w(\mathbf{x}); \theta_{\zeta}); \theta_{\kappa}); \mathbf{x}'; \theta_g). \quad (12)$$

Performance degradation (PD) prediction attempts to forecast the degradation processes in the long term accurately [32]. Compared to the RUL prediction, the PD prediction offers the degradation evolution process from now until a time point in the future, considering it as a sequence prediction. The PD prediction provides supplementary degradation information from a different perspective.

We use health index (HI) $h \in [0, 1]$ to denote the performance degradation status for the aero-engine. The smaller the h , the worse the health condition of the aero-engine. When the aero-engine fails, $h = 0$. The PD prediction results at time t can be denoted as $\hat{h}$ ($\hat{h} = \hat{h}_{t+1:t+lp}$). Similar to RUL prediction, the task-specified layers constitute a PD predictor $\mathcal{H}$, and the PD prediction results $\hat{h}$ are gained in a generative inference manner, as:

$$\hat{h} = \mathcal{H}(\mathcal{G}(\mathcal{K}(\mathcal{Z}(w(\mathbf{x}); \theta_{\zeta}); \theta_{\kappa}); \mathbf{x}'; \theta_g); \theta_h), \quad (13)$$

where θ_h is the parameters in the PD predictor $\mathcal{H}$.

We adopt a piecewise quadratic function Eq. (14) to characterize the performance degradation. It assumes that the performance degradation status keeps the same before the incipient failure (t_c), and then, gradually decreases until the final failure of the engines (T_n). $0 \leq h_t^n \leq 1$, s.t. when the aero-engine n is in perfect health $h_t^n = 1$, and when a failure occurs: $h_t^n = 0$.

$$h_t^n = \begin{cases} 1, & t \leq t_f, \\ 1 - \frac{(t-t_f)^2}{(T_n-t_f)^2}, & \text{else.} \end{cases} \quad (14)$$

Using $\mathcal{L}_h$ to denote the prediction error of the PD performance task ($\mathcal{L}_h(\hat{h}, h)$), the parameters are optimized to achieve:

$$\begin{aligned} \hat{\theta}_h &= \arg \min_{\theta_h} \mathcal{L}_h(\mathcal{H}(\mathcal{G}(\mathcal{K}(\mathcal{Z}(w(\mathbf{x}); \hat{\theta}_{\zeta}); \hat{\theta}_{\kappa}); \mathbf{x}'; \hat{\theta}_g); \theta_h), h), \\ \hat{\theta}_g &= \arg \min_{\theta_g} \mathcal{L}_h(\mathcal{H}(\mathcal{G}(\mathcal{K}(\mathcal{Z}(w(\mathbf{x}); \hat{\theta}_{\zeta}); \hat{\theta}_{\kappa}); \mathbf{x}'; \theta_g); \hat{\theta}_h), h), \\ \hat{\theta}_{\kappa} &= \arg \min_{\theta_{\kappa}} \mathcal{L}_h(\mathcal{H}(\mathcal{G}(\mathcal{K}(\mathcal{Z}(w(\mathbf{x}); \hat{\theta}_{\zeta}); \theta_{\kappa}); \mathbf{x}'; \hat{\theta}_g); \hat{\theta}_h), h), \\ \hat{\theta}_{\zeta} &= \arg \min_{\theta_{\zeta}} \mathcal{L}_h(\mathcal{H}(\mathcal{G}(\mathcal{K}(\mathcal{Z}(w(\mathbf{x}); \theta_{\zeta}); \hat{\theta}_{\kappa}); \mathbf{x}'; \hat{\theta}_g); \hat{\theta}_h), h). \end{aligned} \quad (15)$$

3.2. Loss function and network optimization

All parameters in the MTLTrans network are jointly optimized to simultaneously improve task generalization and individual model performance. The parameters in the shared blocks are shared network parameters, and they are updated to reflect the common knowledge of the degradation of the aero-engine. Meanwhile, parameters in task-specific blocks learn independent information corresponding to the individual tasks, i.e., SOH assessment, RUL estimation and PD prediction. The loss function of each task affects both the individual task and the shared parameters.

The target RUL variable $\hat{y}_i$ is an interval type, the SOH assessment result $\hat{o}_i$ is an ordinal type target with three labels, and HI of PD prediction $\hat{h}_i$ is a vector type. Thus, different kinds of loss functions are chosen and combined in MTLTrans.

As the SOH assessment task is a multi-class classification problem, *Softmax* is used to compute the assessment results and the gradient. The focal cross-entropy loss [33] is employed in the SOH assessment task (Task A), as:

$$\mathcal{L}_o = -\frac{1}{N_S} \sum_{i=1}^{N_S} \sum_{m=1}^M o_{i,m} (1 - p_{i,m})^{\gamma} \log p_{i,m}, \quad (16)$$

where $M = 3$ is the number of health states. o_i is the one-hot representation of the ground truth of the SOH label for the aero-engine at time t (Eq. (17)), and correspondingly, the probability that the i th aero-engine will be evaluated with SOH label m as $p_{i,m}$. $(1 - p_{i,m})^{\gamma}$ is a modulating factor to the cross-entropy loss, with tenable focusing parameter $\gamma \geq 0$. The focal loss reduces the loss contribution from easy-classified examples and extends the range in which an example receives low loss.

$$o_{i,m} = \begin{cases} 1, & o_i = m, \\ 0, & \text{otherwise.} \end{cases} \quad (17)$$

In terms of RUL prediction and PD prediction, they are regression problems, thus we apply the mean square error (MSE) as the loss function (seen in Eqs. (18) and (19)).

$$\mathcal{L}_y = \frac{1}{N_S} \sum_{i=1}^{N_S} (y_i - \hat{y}_i)^2 = \frac{1}{N_S} \sum_{i=1}^{N_S} (y_i - \mathcal{Y}(\mathcal{K}(\mathcal{Z}(w(\mathbf{x}_i))))^2, \quad (18)$$

$$\mathcal{L}_h = \frac{1}{N_S} \sum_{i=1}^{N_S} (h_i - \hat{h}_i)^2 = \frac{1}{N_S} \sum_{i=1}^{N_S} (h_i - \mathcal{H}(\mathcal{G}(\mathcal{K}(\mathcal{Z}(w(\mathbf{x}_i))))^2. \quad (19)$$

Note that, the RUL prediction is a single value $\hat{y}_i$, whereas PD prediction is a time series (vector) $\hat{h}_i = \hat{h}_{i,t+1:t+lp} = \{\hat{h}_{i,t+1}, \dots, \hat{h}_{i,t+lp}\} \in \mathbb{R}^{lp}$. The loss determines how to descend the gradient during training for better generalization performance. We design the overall loss $\mathcal{L}$ of MTLTrans as follows:

$$\mathcal{L} = \alpha_o \mathcal{L}_o + \alpha_y \mathcal{L}_y + \alpha_h \mathcal{L}_h, \quad (20)$$

$$\mathcal{L}(\theta_h, \theta_g, \theta_y, \theta_{\kappa}, \theta_o, \theta_{\zeta}) = \alpha_o \mathcal{L}_o(\theta_o, \theta_{\zeta}) + \alpha_y \mathcal{L}_y(\theta_y, \theta_{\kappa}, \theta_{\zeta}) + \alpha_h \mathcal{L}_h(\theta_h, \theta_g, \theta_{\kappa}, \theta_{\zeta}), \quad (21)$$

where α_o , α_y and α_h are the trade-off parameters with values greater than zero. Note that a significant challenge in MTL stems from the

optimization procedure itself. Balancing the joint learning of each task is essential for an MTL model to avoid the domination impact of a particular task.

Then, we optimize the loss function $\mathcal{L}$ by searching for a saddle point solution of the minimization problem below:

$$\begin{aligned} (\hat{\theta}_o, \hat{\theta}_\zeta) &= \arg \min_{\theta_o, \theta_\zeta} \mathcal{L}(\hat{\theta}_h, \hat{\theta}_g, \hat{\theta}_y, \hat{\theta}_\kappa, \theta_o, \theta_\zeta), \\ (\hat{\theta}_y, \hat{\theta}_\kappa) &= \arg \min_{\theta_y, \theta_\kappa} \mathcal{L}(\hat{\theta}_h, \hat{\theta}_g, \theta_y, \theta_\kappa, \hat{\theta}_o, \hat{\theta}_\zeta), \\ (\hat{\theta}_h, \hat{\theta}_g) &= \arg \min_{\theta_h, \theta_g} \mathcal{L}(\theta_h, \theta_g, \hat{\theta}_y, \hat{\theta}_\kappa, \hat{\theta}_o, \hat{\theta}_\zeta). \end{aligned} \quad (22)$$

Eq. (23) clarifies the optimization of parameters in each function.

$$\begin{aligned} \hat{\theta}_h &= \arg \min_{\theta_h} \mathcal{L}_h(\theta_h, \hat{\theta}_g, \hat{\theta}_\kappa, \hat{\theta}_\zeta) \\ \hat{\theta}_g &= \arg \min_{\theta_g} \mathcal{L}_h(\hat{\theta}_h, \theta_g, \hat{\theta}_\kappa, \hat{\theta}_\zeta) \\ \hat{\theta}_y &= \arg \min_{\theta_y} \mathcal{L}_y(\hat{\theta}_\zeta, \hat{\theta}_\kappa, \theta_y) \\ \hat{\theta}_\kappa &= \arg \{ \min_{\theta_\kappa} \mathcal{L}_y(\hat{\theta}_\zeta, \theta_\kappa, \hat{\theta}_y), \min_{\theta_\kappa} \mathcal{L}_h(\hat{\theta}_\zeta, \theta_\kappa, \hat{\theta}_g, \hat{\theta}_h) \} \\ \hat{\theta}_o &= \arg \min_{\theta_o} \mathcal{L}_o(\hat{\theta}_\zeta, \theta_o) \\ \hat{\theta}_\zeta &= \arg \{ \min_{\theta_\zeta} \mathcal{L}_o(\theta_\zeta, \hat{\theta}_o), \min_{\theta_\zeta} \mathcal{L}_y(\theta_\zeta, \hat{\theta}_\kappa, \hat{\theta}_y), \min_{\theta_\zeta} \mathcal{L}_h(\theta_\zeta, \hat{\theta}_\kappa, \hat{\theta}_g, \hat{\theta}_h) \} \end{aligned} \quad (23)$$

Finally, the Adam algorithm [34] is adopted to train the MTLTrans, and the back-propagation algorithm is utilized to update the parameters in one training step as follows:

$$\begin{aligned} \theta_\zeta &\leftarrow \theta_\zeta - \eta(\alpha_o \frac{\partial \mathcal{L}_o}{\partial \theta_\zeta} + \alpha_y \frac{\partial \mathcal{L}_y}{\partial \theta_\zeta} + \alpha_h \frac{\partial \mathcal{L}_h}{\partial \theta_\zeta}), \\ \theta_o &\leftarrow \theta_o - \eta \alpha_o \frac{\partial \mathcal{L}_o}{\partial \theta_o}, \\ \theta_\kappa &\leftarrow \theta_\kappa - \eta(\alpha_y \frac{\partial \mathcal{L}_y}{\partial \theta_\kappa} + \alpha_h \frac{\partial \mathcal{L}_h}{\partial \theta_\kappa}), \\ \theta_y &\leftarrow \theta_y - \eta \alpha_y \frac{\partial \mathcal{L}_y}{\partial \theta_y}, \\ \theta_g &\leftarrow \theta_g - \eta \alpha_h \frac{\partial \mathcal{L}_h}{\partial \theta_g}, \\ \theta_h &\leftarrow \theta_h - \eta \alpha_h \frac{\partial \mathcal{L}_h}{\partial \theta_h}, \end{aligned} \quad (24)$$

where η is the learning rate. As mentioned earlier, all parameters in MTLTrans are jointly optimized to improve task generalization and individual model performance simultaneously. The loss function of each task affects both the individual task and the shared parameters.

4. Experiments and discussion

This section presents the results of our proposed methods MTLTrans applied to the benchmark datasets C-MAPSS [5]. We trained our models on one machine with NVIDIA TITAN XP GPUs via Python 3.8 with Pytorch 1.9.0. All experiments were repeated ten times, and we reported the average results. Concretely, in this section, we answer the following questions:

- Q1:** How is the performance of the three tasks, i.e., SOH assessment, RUL prediction and PD prediction tasks?
- Q2:** How is the performance of MTLTrans compared with state-of-the-art methods under working-condition shift scenarios?
- Q3:** How to set the trade-off parameters for each task in the MTLTrans? – Parameter Sensitivity Analysis.
- Q4:** Is each block or strategy of MTLTrans conducive to the prediction performance? – Ablation Study.

Datasets Description: We consider the simulated turbofan engine data C-MAPSS [5]. Fig. 4 and Table 2 show the sensors in the turbofan of C-MAPSS. The trajectory in the training set is the RtF trajectory

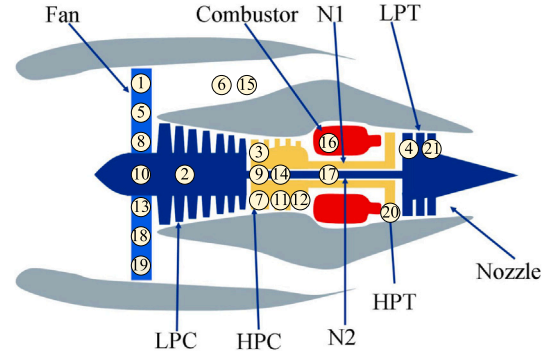


Fig. 4. The sensors in the turbofan of C-MAPSS [5].

Table 2
The description of sensors in turbofan of C-MAPSS [5].

Sensor	Symbol	Description
#1	T2	Total temperature at fan inlet
#2	T24	Total temperature at LPC outlet
#3	T30	Total temperature at HPC outlet
#4	T50	Total temperature at LPT outlet
#5	P2	Pressure at fan inlet
#6	P15	Total pressure in bypass-duct
#7	P30	Total pressure in HPC outlet
#8	Nf	Physical fan speed
#9	Nc	Physical core speed
#10	epr	Engine pressure ratio
#11	Ps30	Static pressure at HPC outlet
#12	phi	Ratio of fuel flow to Ps30
#13	NRf	Corrected fan speed
#14	NRc	Corrected core speed
#15	BPR	Bypass ratio
#16	farB	Burner fuel-air ratio
#17	htBleed	Bleed enthalpy
#18	Nf_dm	Demand fan speed
#19	PCNfr_dmd	Demand corrected fan speed
#20	W31	HPT coolant bleed
#21	W32	LPT coolant bleed

simulated over the entire life of engines, and the trajectory in the test set is the up-to-date (UtD) trajectory. The initial degradation levels and the process noises are set randomly in the simulation. Table 3 offers detailed information on C-MAPSS. In FD001 and FD002, C-MAPSS simulated the fault mode of performance degradation of high-pressure compressors (HPC fault). In contrast, in FD003 and FD004, C-MAPSS simulated the fault mode of performance degradation of high-pressure compressors (HPC fault) and the fault of performance degradation of the fan (Fan fault). In terms of operating conditions, three fundamental parameters determine the operating condition of the aero-engine, i.e., the altitude (H), the Mach numbers (Ma) and the throttle-resolver angle (TRA). In FD001 and FD003, an ideal single operating condition (H≈0 kft, Ma≈0, TRA=100 deg) has been simulated, where kft and deg are for thousand feet and degrees. As for FD002 and FD004, six operating conditions are simulated, i.e., (1) H≈0 kft, Ma≈0, TRA=100 deg; (2) H≈10 kft, Ma≈0.25, TRA=100 deg; (3) H≈20 kft, Ma≈0.7, TRA=100 deg; (4) H≈25 kft, Ma≈0.62, TRA=60 deg; (5) H≈35 kft, Ma≈0.84, TRA=100 deg; (6) H≈42 kft, Ma≈0.84, TRA=100 deg. Operating conditions and fault modes determine the working condition of the aero-engines in each sub-dataset. It should be pointed out that, to simulate the absence of target domain data, the target UtD data is only available at the last time window for each engine in the test datasets. That means the target data can only be used in the testing phase, not training phase. Similar to other works in the literature [2], we use a piece-wise linear degradation model to define the correct RUL values in the training and test datasets, and the maximum RUL y_{max} is set to be 125.

Table 3
Description of C-MAPSS [5].

Dataset	FD001	FD002	FD003	FD004
Training trajectories	100	260	100	249
Test trajectories	100	259	100	248
Maximum lifespan (cycles)	362	378	525	543
Average lifespan (cycles)	206	206	247	245
Minimum lifespan (cycles)	128	128	145	128
Operating conditions	1	6	1	6
Fault modes	1	1	2	2

Data Distribution in C-MAPSS: Fig. 5 illustrates the distribution of eight randomly selected sensor signals in the sub-datasets of C-MAPSS. We can see that sensory data in FD001 and FD003 show a similar distribution in sensor 3 and sensor 14. Data distributions vary from each other, and the discrepancy is evident in sensor 11, sensor 21, and sensor 15. In contrast, data distributions of sensor 7 of the four sub-datasets are inconsistent due to the varying fault modes. On the other hand, we observe a lower distribution shift between the dataset pair of FD002 and FD004, seen in Figs. 5(c) and (f). There are multiple bell curves for this pair's distribution of sensory data. We can conjecture that this is due to the influence of the six different operating conditions. In practice, these distribution shifts make models data specific, i.e., a model trained in one dataset often does not perform well in another dataset subjected to different data distribution. Furthermore, data distribution in sensor 5 is attractive since there is only a single value of several values. These signals are excluded from the RUL model because they do not contain degradation information. In general, we exclude the sensors with highly meaningless information (consistent or periodical readings), and 14 sensors (No. 2–4, 7–9, 11–15, 17, 20, and 21, are considered in the following experiment.

12 Working-condition Shift Scenarios: As shown in Fig. 5, all these four sub-datasets of C-MAPSS are in different working conditions (operating settings or fault modes). Here, we adopt FD001→FD002 to denote the working-condition shift scenario from FD001 to FD002, which means using the training data (RtF trajectories) in FD001 as the source domain (training domain) and test data (UtD trajectories) in FD002). In this way, we construct 12 working-condition shift scenarios as (a): FD001→FD002, (b): FD001→FD003, (c): FD001→FD004, (d): FD002→FD001, (e): FD002→FD003, (f): FD002→FD004, (g): FD003→FD001, (h): FD003→FD002, (i): FD003→FD004, (j): FD004→FD001, (k): FD004→FD002, (l): FD004→FD003. Table 4 describes the 12 cross-working-condition scenarios of C-MAPSS, in which OC and FM denote the operating condition and fault mode. We can also observe that FD001→FD002 is a relatively easy scenario of working-condition shift compared with FD001→FD004. Because there is only the operating-condition shift in the former (FD001→FD002), whereas two kinds of working-condition shifts, i.e., the operating-condition shift and fault-modes shift, exist in the latter (FD001→FD004). The data distribution discrepancy of FD001→FD004 is more distinct than that of FD001→FD002, as shown in Fig. 5.

Details of Parameter Setting: Our model follows the design of Informer [29], and contains a 3-layer stack and a 1-layer stack (1/4 input) in the encoder, and a 2-layer decoder. The learning rate η starts from $5e^4$, decaying twice smaller every epoch. The input RtF trajectories have been standard normalized first [2]. Note that, with the absence of target data, no specified parameter turning or domain adaption process is accomplished in our model MTLTrans. Hence, we only configure one structure for the four datasets with 12 cross-working-condition scenarios. Compared with the previous study, our model is more flexible and practical with the participation of target data in the model training or updating process. For more details, one can refer to the source code at <https://github.com/ZhiyaoZhang12/MTLTrans.git>. More details about the parameter settings are shown in Table 5.

Performance Metrics: The scoring function (Eq. (25)) [5] and the root mean square error (RMSE) (Eq. (26)) are adopted to evaluate

Table 4
The description of 12 working-condition shift scenarios.

Index	Scenarios	Shift description	OC shift		FM shift	
			1→6	6→1	1→2	2→1
1	FD001→FD002	1OC1FM→6OC1FM	✓			
2	FD001→FD003	1OC1FM→1OC2FM			✓	
3	FD001→FD004	1OC1FM→6OC2FM	✓		✓	
4	FD002→FD001	6OC1FM→1OC1FM		✓		
5	FD002→FD003	6OC1FM→1OC2FM		✓	✓	
6	FD002→FD004	6OC1FM→6OC2FM			✓	
7	FD003→FD001	1OC2FM→1OC1FM				✓
8	FD003→FD002	1OC2FM→6OC1FM	✓			✓
9	FD003→FD004	1OC2FM→6OC2FM	✓			
10	FD004→FD001	6OC2FM→1OC1FM		✓		✓
11	FD004→FD002	6OC2FM→6OC1FM				
12	FD004→FD003	6OC2FM→1OC2FM		✓		✓

Table 5
The parameter setting in MTLTrans.

Parameters	Values
window size	36
learning rate η	0.00005
patience	3
batch size	128
maximum RUL y_{max}	125 [16,25]
the incipient failure t_e	when $y_i = 124$
severe failure threshold t_e	when $h_i = 0.5$
dropout rate	0.05
dimension of the main structure d_{model}	512
number of heads of attention n_{head}	8
sampling factor	5
trade-offs of three tasks $\alpha_o, \alpha_y, \alpha_h$	20,0.005,1
γ in forcal loss	2
PD prediction horizon lp	16
length of the masked sequence lm in $\mathcal{H}$	16
number of FCLs in $\mathcal{O}$	2
number of FCLs in $\mathcal{K}$	1
number of FCLs in $\mathcal{Y}$	3
number of FCLs in $\mathcal{H}$	1
number of neurons in $\mathcal{O}$	512,128,3
number of neurons in $\mathcal{K}$	512,256
number of neurons in $\mathcal{Y}$	256,128,64,1
number of neurons in $\mathcal{H}$	512,1

the performance of RUL prediction methods. $e_i = \hat{y}_i - y_i$ denotes the prediction error between the predicted RUL $\hat{y}_i$ and the ground truth y_i . The Score increases exponentially with the RUL prediction error. Besides, the scoring function is asymmetric to give more penalties for late predictions than the early predictions, which is in line with the risk aversion attitude of highly hazardous industries like aviation, aerospace, nuclear and chemical industries.

$$Score = \begin{cases} \sum_{i=1}^{N_T} (\exp(-\frac{e_i}{13}) - 1), & e_i \leq 0 \\ \sum_{i=1}^{N_T} (\exp(\frac{e_i}{10}) - 1), & else \end{cases} \quad (25)$$

$$RMSE = \sqrt{\frac{1}{N_T} \sum_{i=1}^{N_T} (e_i)^2} \quad (26)$$

Q1: How is the performance of the three tasks?

The SOH assessment task focuses on the present, whereas the PD prediction task is long-term-oriented. These two auxiliary tasks facilitate the performance of the main task (i.e., the RUL prediction task) from different perspectives. Besides, in the O&M of aero-engine, the results of the three tasks can provide the manager with comprehensive health information about the aero-engine from different perspectives, contributing significantly to a reliable and cost-effective (predictive maintenance) PdM decision.

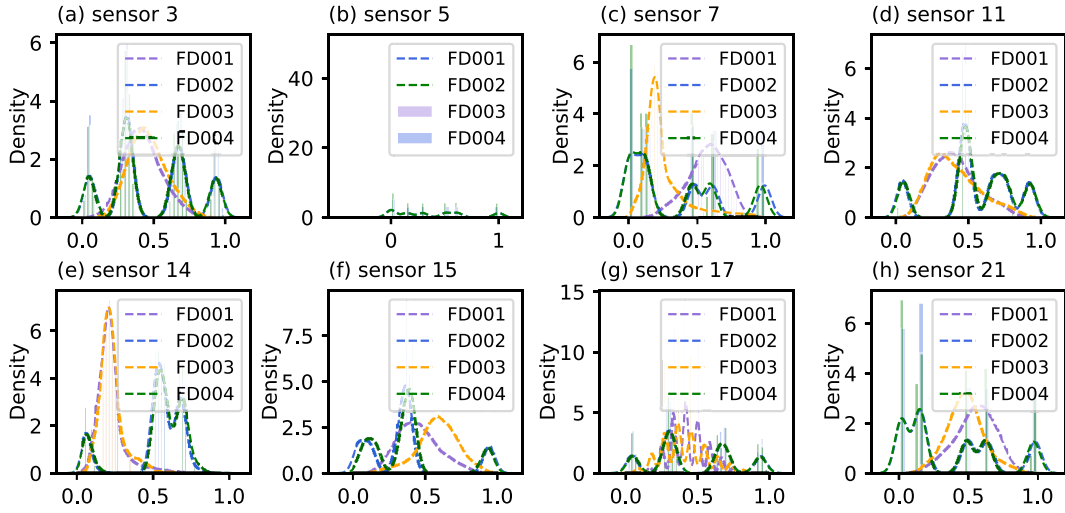


Fig. 5. The sensor data distribution of four sub-datasets in C-MAPSS: (a) sensor 3, (b) sensor 5, (c) sensor 7, (d) sensor 11, (e) sensor 14, (f) sensor 15, (g) sensor 17, (h) sensor 21.

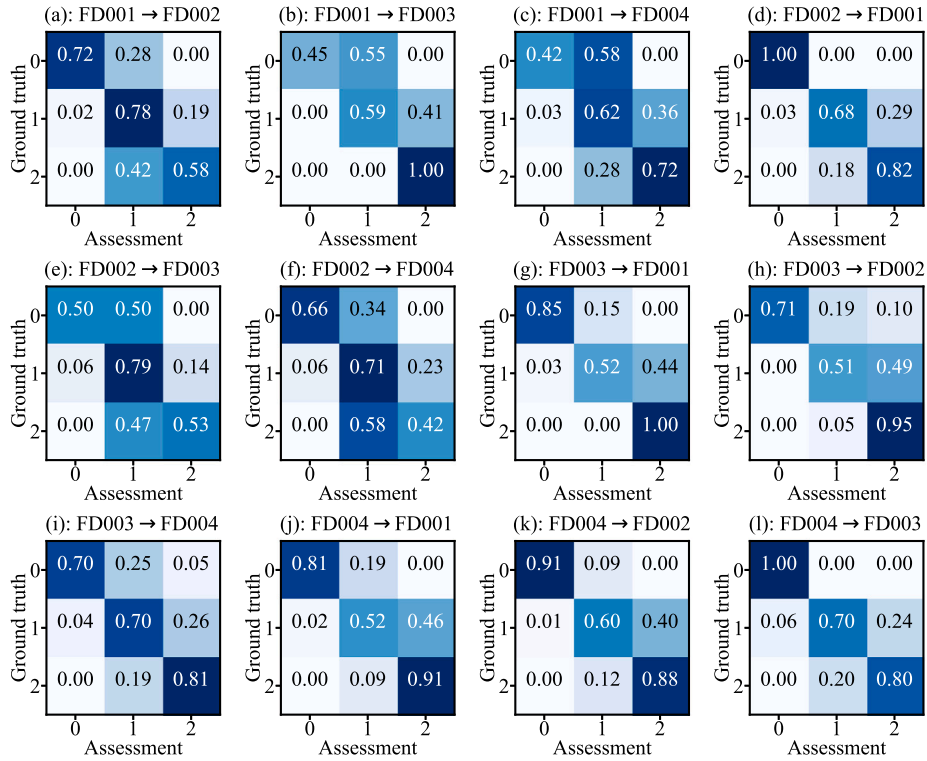


Fig. 6. The confusion matrixes for the SOH assessment: (a) FD001→FD002, (b) FD001→FD003, (c) FD001→FD004, (d) FD002→FD001, (e) FD002→FD003, (f) FD002→FD004, (g) FD003→FD001, (h) FD003→FD002, (i) FD003→FD004, (j) FD004→FD001, (k) FD004→FD002, (l) FD004→FD003.

Task A: SOH Assessment: Fig. 6 provides the performance of MTLTrans on the SOH assessment task. We observe that distinguishing between the initial degradation stage (class 1) and the severe degradation stage (class 0) is less challenging than that between the initial degradation stage (class 1) and the health stage (class 2), for example, FD001→FD003, FD001→FD004 and FD002→FD003. This can be attributed to the weak degradation features at the beginning of the initial degradation stage. In general, the MTLTrans produces acceptable assessment results in most working-condition shift scenarios. Intuitively, the SOH assessment task can be efficiently implemented under the working-condition shift scenarios under the MTL structure, posing potential benefits to other integrated learning tasks, i.e., RUL prediction and PD prediction.

Task B: RUL Prediction: Fig. 7 shows the predicted RULs of aero-engines in the last cycle (100, 259, 100 and 248 aero-engines in FD001, FD002, FD003 and FD004, respectively) of each target sub-dataset. To better observe the estimation results, the order of aero-engines is rearranged descendingly by RUL values. From Fig. 7, we can observe that: (1) In general, the predicted RULs follow the ground truth, verifying the prediction efficiency of the proposed MTLTrans. (2) The MTLTrans gains excellent performance, particularly when a failure is likely to occur, benefiting from the distinct degradation features during the severe degradation stage. Last, (3) in some working-condition scenarios, i.e., FD002→FD001, FD004→FD003, FD004→FD002, the prediction performance is remarkable, owing to the single working-condition shift

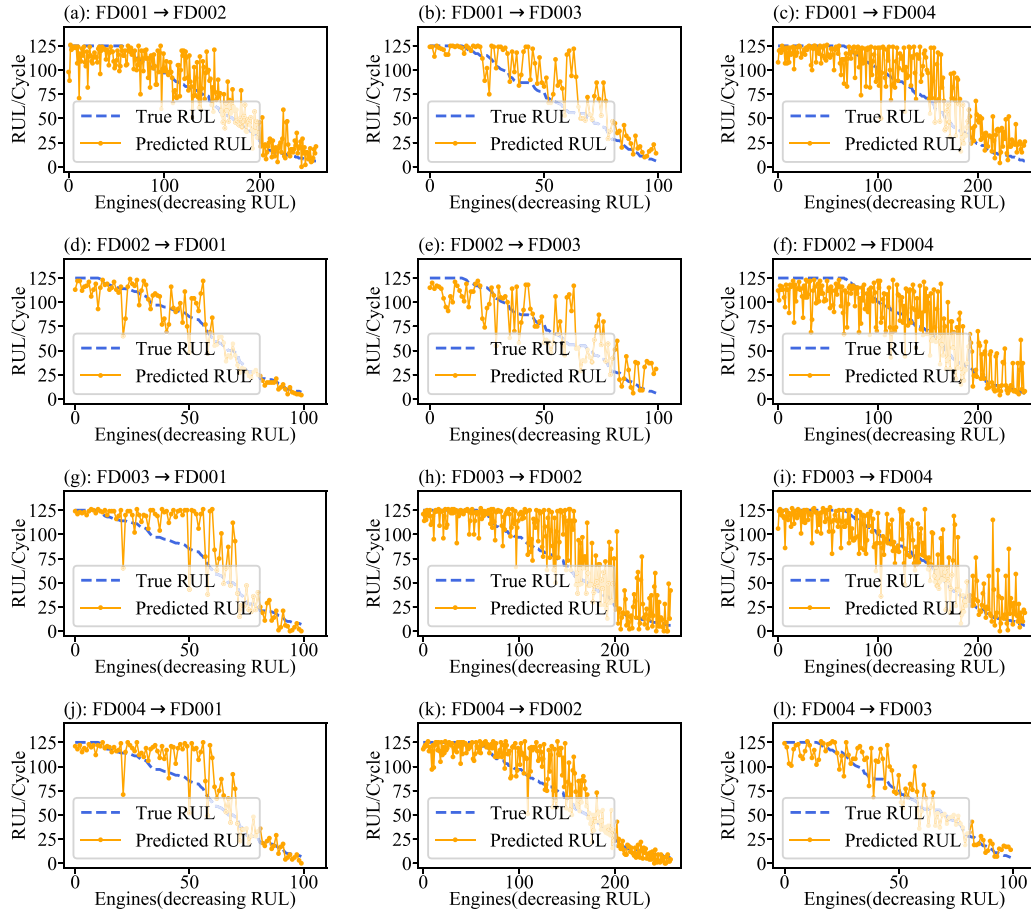


Fig. 7. The ground truth (blue) and predicted RULs (orange): (a) FD001→FD002, (b) FD001→FD003, (c) FD001→FD004, (d) FD002→FD001, (e) FD002→FD003, (f) FD002→FD004, (g) FD003→FD001, (h) FD003→FD002, (i) FD003→FD004, (j) FD004→FD001, (k) FD004→FD002, (l) FD004→FD003. To better observe the estimation results, the order of aero-engines is rearranged in descending order by RUL values. There are 100, 259, 100 and 248 aero-engines in FD001, FD002, FD003 and FD004, respectively.

(operating-condition shift or fault-mode shift) from a complex one to a simple one.

To further investigate the generalization of the MTLTrans, we made additional experiments about the RUL predictions for the entire life cycle of the aero-engine in the target domain dataset. We choose four representative scenarios of working-condition shifts, i.e., FD001→FD002 (OP shift), FD002→FD003 (OP shift and FM shift), FD003→FD001 (FM shift), FD004→FD001 (OP shift and FM shift). Fig. 8 describes the RUL prediction performance for the randomly selected aero-engine units. For these 12 aero-engines with different lengths of trajectories, the MTLTrans yield desirable RUL prediction results close to the true RUL among the life cycles. It demonstrates the excellent generalization ability of MTLTrans in cross-working conditions.

Task C: PD Prediction: Fig. 9 illustrates the performance of the PD prediction of several randomly selected aero-engines among the 12 working-condition shift scenarios. It can be observed that the PD state of most of the aero-engines can be predicted efficiently since the gap between the true HIs and the predicted HIs is acceptable in most working-condition shift scenarios. The PD prediction in our study with $l_p = 16$ is a relatively long-term prediction that relies heavily on the efficient extraction of long-term temporary characteristics.

Q2: How is the performance of MTLTrans compared with state-of-the-art methods under working-condition shift scenarios?

Tables 6 and 7 summarize the results of the proposed MTLTrans for RUL prediction of the C-MAPSS datasets with classical deep neural networks, i.e., deep neural networks (DNN), deep CNN (DCNN) and

BiLSTM without domain adaption, and six state-of-the-art methods that appeared in the literature over the last three years. The best results are highlighted in boldface, and the second-best results are underlined. The different performances between RMSE and Score are due to the unbalanced punishment of Score on the late prediction with respect to the early prediction.

Tables 6 and 7 show that: (1) The proposed MTLTrans achieves the best performance on RMSE (19.22) by decreasing 18.83% on average. In the working-condition-shift scenarios of FD001→FD002, FD001→FD004, and FD002→FD004, the performance improvement of MTLTrans on RMSE is significant, by decreasing 14.81%, 24.21% and 10.24% than the second-best performance of each scenario, respectively. We attribute the superior performance to the MTL structure of the MTLTrans, which provides generalization results with the help of SOH assessment and PD prediction tasks. (2) In contrast, when the working-condition shift scenarios occur from two-fault mode to the single fault mode, such as FD003→FD001/FD002, the MTLTrans performs inferior to CADA [6] and DIDRLSTM [15], in which two kinds of learning strategy have been integrated. For instance, CADA [6] combines contrastive learning and adversarial learning, whereas DIDRLSTM [15] combines statistical- and adversarial-based DA, resulting in competitive results in the operational-condition shift problem and the fault-mode shift problem. (3) Interestingly, the second-best performance on RMSE has been yielded by the BiLSTM, in which no DA technology has been employed. Compared with the DA approaches, i.e., DAN [10], JAN [11] and DANN [24], the BiLSTM performs superior in the working-condition shift scenarios of FD001→FD004 and FD002→FD004. The negative transfer may happen in these DA methods under the fault-mode shift scenario of 2FM→1FM.

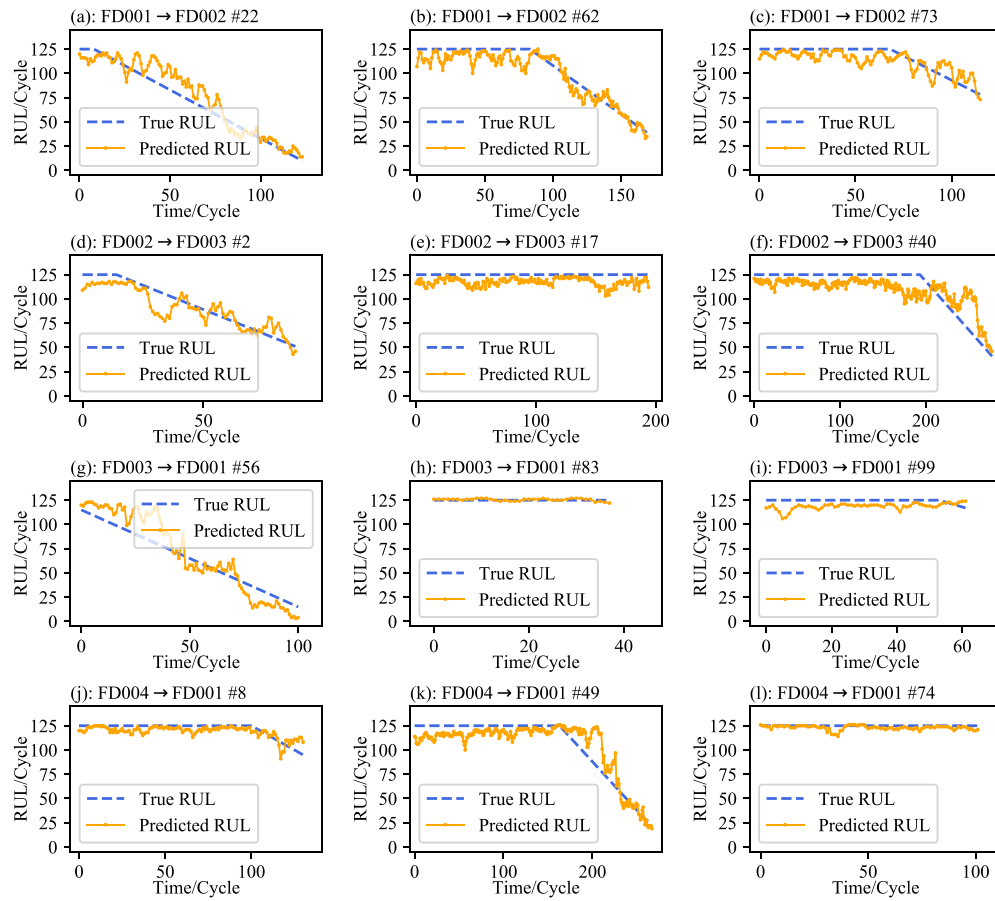


Fig. 8. The RUL prediction performance for the random selected scenarios and aero-engine units: (a) FD001→FD002 unit 22, (b) FD001→FD002 unit 62, (c) FD001→FD002 unit 73, (d) FD002→FD003 unit 2, (e) FD002→FD003 unit 17, (f) FD002→FD003 unit 40, (g) FD003→FD001 unit 56, (h) FD003→FD001 unit 83, (i) FD003→FD001 unit 99, (j) FD004→FD001 unit 8, (k) FD004→FD001 unit 49, (l) FD004→FD001 unit 74. To better observe the estimation results, the order of aero-engines is rearranged in descending order by RUL values. There are 100, 259, 100 and 248 aero-engines in FD001, FD002, FD003 and FD004, respectively.

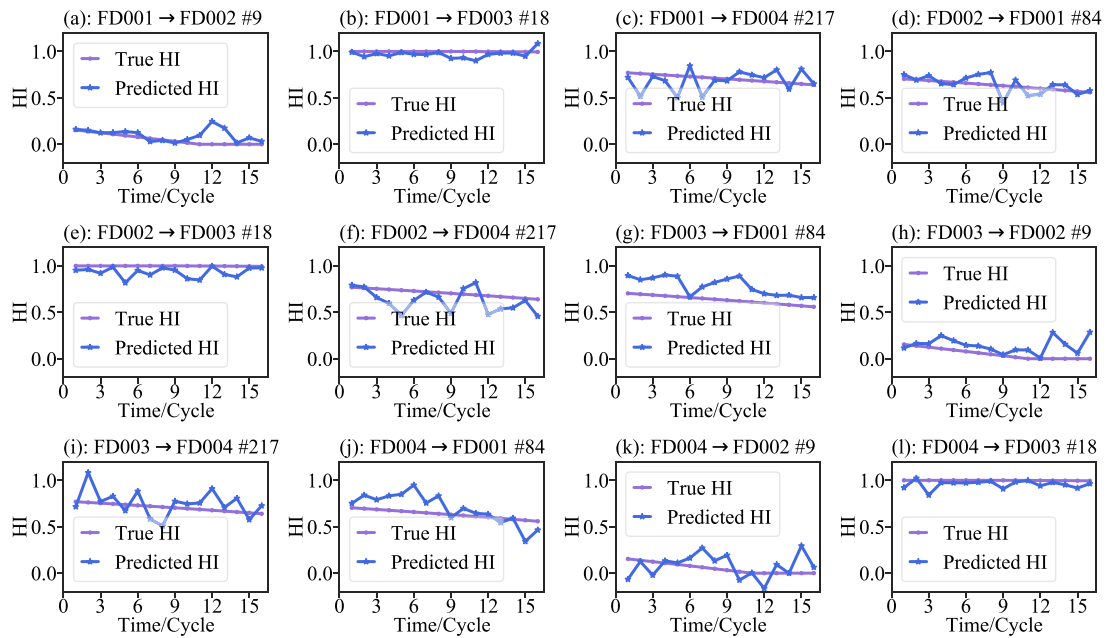


Fig. 9. The PD prediction performance: (a) FD001→FD002 #9, (b) FD001→FD003 #18, (c) FD001→FD004 #217, (d) FD002→FD001 #84, (e) FD002→FD003 #18, (f) FD002→FD004 #217, (g) FD003→FD001 #84, (h) FD003→FD002 #9, (i) FD003→FD004 #217, (j) FD004→FD001 #84, (k) FD004→FD002 #9, (l) FD004→FD003 #18.

Table 6

RUL prediction performance comparison on RMSE of MTLTrans with state-of-the-art methods on C-MAPSS.

Source	Target	DNN	DCNN	BiLSTM	DAN [10]	JAN [11]	DDC [35]	DANN [24]	CADA [6]	DIDRLSTM [15]	MTLTrans
FD001	FD002	24.66	34.44	24.70	35.9	35.7	32.6	32.4	<u>19.52</u>	29.4	16.63
	FD003	25.56	24.33	23.14	24.4	25.4	21.5	23.7	39.58	<u>20.9</u>	20.83
	FD004	29.74	34.74	<u>28.71</u>	38.7	38.8	36.8	36.6	31.23	34.4	21.76
FD002	FD001	16.43	17.46	14.50	29.9	20.1	20.3	21.6	<u>13.88</u>	16.1	13.67
	FD003	24.34	23.13	25.01	24.8	23.7	22.1	24.7	33.53	<u>21.1</u>	20.60
	FD004	26.69	<u>24.71</u>	24.78	37.3	38	37.1	37.5	33.71	34	22.18
FD003	FD001	21.16	29.54	22.20	<u>19.2</u>	23.5	22.5	23.9	19.54	18.8	21.22
	FD002	31.14	30.30	32.09	34.9	35.4	33.2	34.4	19.33	29.4	<u>23.79</u>
	FD004	30.52	24.56	29.64	33.8	33	32.7	33.1	20.61	32.3	<u>21.04</u>
FD004	FD001	<u>19.71</u>	29.46	21.74	23.8	24.8	22.9	22.3	20.1	21.3	18.44
	FD002	21.66	25.02	20.63	31.4	32.2	31.2	32.3	<u>18.5</u>	28.9	16.86
	FD003	20.13	15.66	16.95	19	19.8	19.7	19.7	<u>14.49</u>	16.9	13.56
Average		24.35	26.11	<u>23.68</u>	29.43	29.2	27.72	28.52	23.95	25.29	19.22

The bold entities represent the best performance and the underlined ones are the second-best performance.

Table 7

RUL prediction performance comparison on Score of MTLTrans with state-of-the-art methods on C-MAPSS.

Source	Target	DNN	DCNN	BiLSTM	DAN [10]	JAN [11]	DDC [35]	DANN [24]	CADA [6]	DIDRLSTM [15]	MTLTrans
FD001	FD002	33355	149475	41575	11675	15684	7293	6764	<u>2122</u>	4822	1618
	FD003	2476	6225	2283	650	648	457	656	8415	470	2286
	FD004	20945	114306	38509	31083	11865	8124	8240	11577	<u>5538</u>	4714
FD002	FD001	494	846	386	412	573	483	531	351	213	<u>331</u>
	FD003	1988	4668	2842	653	567	383	684	5213	474	1510
	FD004	5836	7300	5594	8977	10204	7034	7759	15106	<u>5471</u>	3860
FD003	FD001	5896	10258	7943	737	<u>628</u>	813	777	1451	442	1890
	FD002	184291	57121	184191	14969	8287	6988	8996	<u>5257</u>	4127	9098
	FD004	179124	80663	154578	5451	4681	4259	4497	3219	<u>4123</u>	13261
FD004	FD001	1362	13937	2973	754	916	784	615	840	553	1096
	FD002	12594	24781	10264	4787	5736	5485	5814	4460	<u>3483</u>	2189
	FD003	2625	551	1296	<u>323</u>	472	466	436	682	249	383
Average		37576	39178	37703	6706	5022	3547	3814	4891	2497	<u>3520</u>

The bold entities represent the best performance and the underlined ones are the second-best performance.

Q3: How to set the trade-off parameters for each task in the MTLTrans? – Parameter sensitivity analysis

To evaluate the impact of key hyperparameters α_o , α_y and α_h in the total loss (Eq. (20)) of the MTLTrans, we implement the parameter sensitivity analysis. These three hyperparameters determine the trade-offs of the three tasks, i.e., the SOH assessment task, the RUL prediction task and the PD prediction task, respectively, in the optimization procedure. Balancing the joint learning of each task through the value of the gradient is essential for an MTL model to avoid the domination impact of a particular task. In essence, blindly increasing the weight of the regression loss of the RUL prediction task (α_y) cannot produce desirable prediction results. When the α_y is relatively great, the training process would be significantly dominated by the loss of the RUL prediction task, resulting in the single task learning and losing the contributions of the auxiliary tasks (the SOH assessment task and the PD prediction task).

We take working-condition shift scenarios FD004→FD001/FD002/FD003 as an example analyze the parameters' sensitivity. Note that, as the losses of the three tasks are of different magnitude, there is no comparability of the values of each trade-off. The performance of different parameter settings is illustrated in Fig. 10, and the best performance on RUL prediction is gained when $\alpha_o = 20$, $\alpha_y = 0.005$ and $\alpha_h = 1$. In FD004→FD003, the results under this set are not optimal but are still competitive with the best performance of the scenario. From Fig. 10, we can observe that: (1) For scenario FD004→FD001 (the green), increasing the value of α_o may improve the RUL prediction performance, but further increasing it degrades performance; (2) For scenario FD004→FD002 (the purple), the three trade-offs show different impacts on the RUL prediction performance; (3) For scenario FD004→FD003 (the blue), the trade-offs of these three tasks slightly impact the training of the MTLTrans.

Q4: Is each block or strategy of MTLTrans conducive to the prediction performance? – Ablation study

We conduct the ablation experiment to explore the contribution of each block or strategy MTLTrans adopted, including the feature distiller $\mathcal{K}$, auxiliary tasks $\mathcal{O}$ and $\mathcal{H}$, Transformer-based decoder $\mathcal{G}$, hierarchy sharing style and the dropout strategy in the RUL predictor $\mathcal{Y}$. In the following, we summarize the details of each case. We summarize the performance of each case in Table 8, where the best results of each working-condition scenario are highlighted.

Case 1: MTLTrans without the feature distiller (wo $\mathcal{K}$).

Case 2: MTLTrans with hard-sharing after the decoder $\mathcal{G}$, instead of the hierarchical sharing.

Case 3: Case 2 and without Transformer-based decoder (wo $\mathcal{G}$).

Case 4: MTLTrans without auxiliary tasks (wo $\mathcal{O}$ and $\mathcal{H}$).

Case 5: MTLTrans without dropout strategy in the RUL predictor (wo dropout in $\mathcal{Y}$).

From Table 8, we can observe that (1) The most inferior performance is gained in Case 4, which demonstrates that the introduction of SOH assessment and PD prediction tasks is the dominant reason for the superior performance of MTLTrans under the scenarios of the working-condition shift. It verifies the potential of MTL methods in the data shift problem. (2) The performance of Case 5 described comparing the results of MTLTrans, because of the reduction of generalization capability without the dropout strategy. (3) Case 1, Case 2 and Case 3 demonstrate the effectiveness of the feature distiller, hierarchical sharing and the encoder–decoder structure. (4) All these five cases yield acceptable results compared with the state-of-the-art methods in Table 6, indicating the efficiency of the Transformer-based network in capturing the high-dimensional and temporal dependence characteristics among multi-sensor time series.

Table 8
Ablation study on MTLTrans.

Source dataset	Target dataset	MTLTrans		Case 1		Case 2		Case 3		Case 4		Case 5	
		Score	RMSE	Score	RMSE	Score	RMSE	Score	RMSE	Score	RMSE	Score	RMSE
FD001	FD002	1618	16.63	1942	19.03	1919	18.09	8134	22.51	2575	18.80	2053	18.20
	FD003	2287	20.83	1564	21.56	2044	20.77	1695	19.29	3204	22.51	2125	21.04
	FD004	4714	21.76	3754	24.05	3859	23.36	18848	27.83	7611	25.10	4456	23.67
FD002	FD001	331	13.68	320	13.18	339	13.30	353	14.19	387	14.79	348	13.49
	FD003	1510	20.60	2026	22.07	1830	20.80	1833	22.45	4143	25.65	1637	21.18
	FD004	3860	22.18	4614	23.16	3803	22.44	7506	27.05	12004	26.99	3860	22.41
FD003	FD001	1890	21.22	1296	19.91	1757	21.05	2582	22.47	4107	24.90	1782	20.90
	FD002	9098	23.79	24983	25.87	33916	27.39	62811	27.74	49351	28.89	31302	27.29
	FD004	13261	21.04	29136	23.42	27929	23.33	66113	25.46	47635	27.29	30606	23.39
FD004	FD001	1096	18.44	1544	19.92	1253	18.88	1472	20.16	1787	20.80	1372	19.45
	FD002	2189	16.86	2596	17.43	2292	17.14	2648	17.45	3428	18.61	2637	17.60
	FD003	383	13.56	369	13.51	320	13.05	569	14.19	500	14.50	345	13.30
Average		3520	19.22	6179	20.26	6772	19.97	14547	21.73	11394	22.40	6877	20.16

The bold entities represent the best performance among all the cases in each working-condition shift scenario.

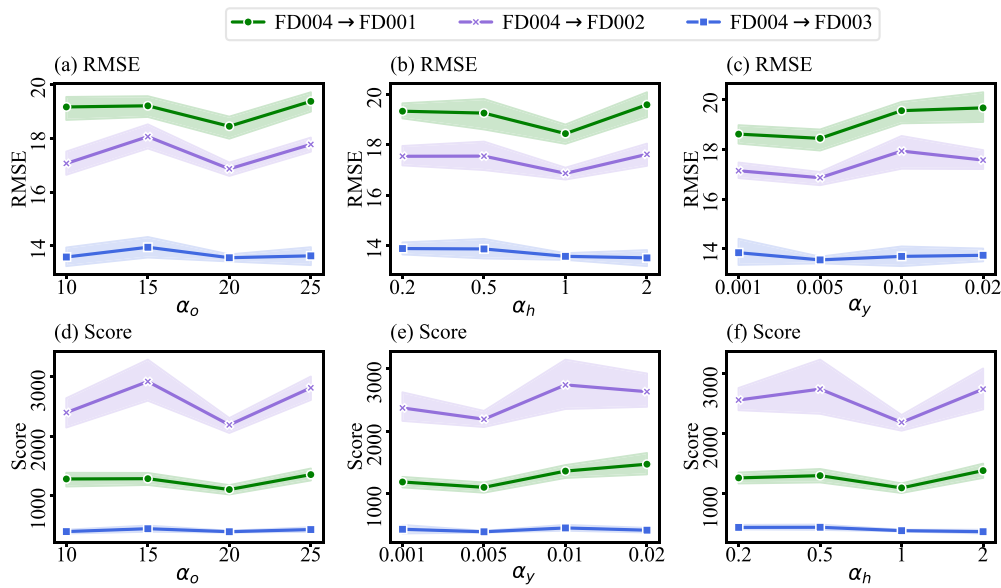


Fig. 10. The results of the sensitive study for the parameters α_o , α_y and α_h , which are the trade-off parameters of the SOH assessment task, the RUL prediction task and the PD prediction task in the MTL model (MTLTrans), respectively.

5. Conclusion

In this work, we have studied the problem of RUL prediction of aero-engines in the absence of target data under working-condition shift scenarios. A Transformer-based multi-task (MTL) learning method (MTLTrans) has been proposed to improve the generalization capability and produce robust results against the data shift. In MTLTrans, we conduct the RUL prediction task along with the SOH assessment task and the PD prediction task, integrating them in a hierarchical sharing manner. The proposed MTLTrans solely relies on the source data in the model training phase, solving the limitation of the data availability of historical degradation time series in shifted working conditions of the target domain. Experiments on the C-MAPSS dataset demonstrate the excellent generalization capability and robustness of MTLTrans in the RUL prediction of aero-engines with the working-condition shift problem. This is the first primary exploration of the MTL method for the RUL prediction under the working-condition shift scenarios. This study demonstrated the potential of the MTL technology facing data shift problems. For future work, we plan to explore the data distribution shift problem, such as the divergence of conditional probability distributions caused by different entities. Also, we will explore the tradeoffs of shift-invariant and shift-related information.

CREdit authorship contribution statement

Zhiyao Zhang: Writing – original draft, Visualization, Validation, Software, Methodology, Conceptualization. **Xiaohui Chen:** Supervision, Project administration, Funding acquisition. **Enrico Zio:** Writing – review & editing, Validation, Supervision, Conceptualization. **Longxiao Li:** Writing – original draft, Methodology, Conceptualization.

Declaration of competing interest

All authors disclosed no conflict of interest.

Data availability

Data will be made available on request.

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References

- [1] Li Y, Shi Y, Zhang Z, Lu N, Wang X, Zio E. Condition-based maintenance for performance degradation under non-periodic unreliable inspections. *IEEE Trans Artif Intell* 2022;1–13. <http://dx.doi.org/10.1109/TAI.2022.3197680>.
- [2] Zhang Z, Chen X, Zio E. A framework for predicting the remaining useful life of machinery working under time-varying operational conditions. *Appl Soft Comput* 2022;126:109164. <http://dx.doi.org/10.1016/j.asoc.2022.109164>.
- [3] Li Y, Wang X, Lu N, Jiang B. Conditional joint distribution-based test selection for fault detection and isolation. *IEEE Trans Cybern* 2021;52(12):13168–80. <http://dx.doi.org/10.1109/TCYB.2021.3105453>.
- [4] Li Y, Xu S, Chen H, Jia L, Ma K. A general degradation process of useful life analysis under unreliable signals for accelerated degradation testing. *IEEE Trans Ind Inf* 2022;1–9. <http://dx.doi.org/10.1109/TII.2022.3224960>.
- [5] Frederick DK, Decastro JA, Litt JS. User's guide for the commercial modular aero-propulsion system simulation (C-MAPSS). Nasa/Arl 2007.
- [6] Ragab M, Chen Z, Wu M, Foo CS, Kwok CK, Yan R, et al. Contrastive adversarial domain adaptation for machine remaining useful life prediction. *IEEE Trans Ind Inf* 2021;17(8):5239–49. <http://dx.doi.org/10.1109/TII.2020.3032690>.
- [7] Zhuang J, Jia M, Zhao X. An adversarial transfer network with supervised metric for remaining useful life prediction of rolling bearing under multiple working conditions. *Reliab Eng Syst Saf* 2022;225:108599. <http://dx.doi.org/10.1016/j.res.2022.108599>.
- [8] Chen J, Li D, Huang R, Chen Z, Li W. Aero-engine remaining useful life prediction method with self-adaptive multimodal data fusion and cluster-ensemble transfer regression. *Reliab Eng Syst Saf* 2023;234:109151. <http://dx.doi.org/10.1016/j.res.2023.109151>.
- [9] Li W, Shang Z, Gao M, Qian S, Feng Z. Remaining useful life prediction based on transfer multi-stage shrinkage attention temporal convolutional network under variable working conditions. *Reliab Eng Syst Saf* 2022;226:108722. <http://dx.doi.org/10.1016/j.res.2022.108722>.
- [10] Jin Y, Qin C, Liu J, Lin K, Liu C. A novel domain adaptive residual network for automatic atrial fibrillation detection. *Knowl-Based Syst* 2020;203:106122. <http://dx.doi.org/10.1016/j.knsys.2020.106122>.
- [11] Zhao K, Jiang H, Wang K, Pei Z. Joint distribution adaptation network with adversarial learning for rolling bearing fault diagnosis. *Knowl-Based Syst* 2021;222:106974. <http://dx.doi.org/10.1016/j.knsys.2021.106974>.
- [12] Xiao Y, Shao H, Han S, Huo Z, Wan J. Novel joint transfer network for unsupervised bearing fault diagnosis from simulation domain to experimental domain. *IEEE/ASME Trans Mechatronics* 2022;27(6):5254–63. <http://dx.doi.org/10.1109/TMECH.2022.3177174>.
- [13] Zhang J, Li X, Tian J, Jiang Y, Luo H, Yin S. A variational local weighted deep sub-domain adaptation network for remaining useful life prediction facing cross-domain condition. *Reliab Eng Syst Saf* 2023;231:108986. <http://dx.doi.org/10.1016/j.res.2022.108986>.
- [14] Shi H, Huang C, Zhang X, Zhao J, Li S. Wasserstein distance based multi-scale adversarial domain adaptation method for remaining useful life prediction. *Appl Intell* 2023;53(3):3622–37. <http://dx.doi.org/10.1007/s10489-022-03670-6>.
- [15] Fu S, Zhang Y, Lin L, Zhao M, Sheng Zhong S. Deep residual LSTM with domain-invariance for remaining useful life prediction across domains. *Reliab Eng Syst Saf* 2021;216:108012. <http://dx.doi.org/10.1016/j.res.2021.108012>.
- [16] Xiao L, Duan F, Tang J, Abbott D. A noise-boosted remaining useful life prediction method for rotating machines under different conditions. *IEEE Trans Instrum Meas* 2021;70(99):1–12. <http://dx.doi.org/10.1109/TIM.2021.3064810>.
- [17] Ding Y, Jia M, Cao Y, Ding P, Zhao X, Lee C-G. Domain generalization via adversarial out-domain augmentation for remaining useful life prediction of bearings under unseen conditions. *Knowl-Based Syst* 2023;261:110199. <http://dx.doi.org/10.1016/j.knsys.2022.110199>.
- [18] Zhuang J, Jia M, Ding Y, Zhao X. Health assessment of rotating equipment with unseen conditions using adversarial domain generalization toward self-supervised regularization learning. *IEEE/ASME Trans Mechatronics* 2022;27(6):4675–85. <http://dx.doi.org/10.1109/TMECH.2022.3163289>.
- [19] Vandenhende S, Georgoulis S, Gansbeke WV, Proesmans M, Dai D, Gool LV. Multi-task learning for dense prediction tasks: A survey. *IEEE Trans Pattern Anal Mach Intell* 2022;44:3614–33. <http://dx.doi.org/10.1109/TPAMI.2021.3054719>.
- [20] Girshick R. Fast R-CNN. In: Proceedings of the IEEE international conference on computer vision. 2015, p. 1440–8. <http://dx.doi.org/10.1109/ICCV.2015.169>.
- [21] Wang H, Lin T, Cui L, Ma B, Dong Z, Song L. Multitask learning-based self-attention encoding atrous convolutional neural network for remaining useful life prediction. *IEEE Trans Instrum Meas* 2022;71:1–8. <http://dx.doi.org/10.1109/TIM.2022.3185312>.
- [22] Kim TS, Sohn SY. Multitask learning for health condition identification and remaining useful life prediction: deep convolutional neural network approach. *J Intell Manuf* 2021;32(8):2169–79. <http://dx.doi.org/10.1007/s10845-020-01630-w>.
- [23] Zhang Y, Xin Y, Liu Z-w, Chi M, Ma G. Health status assessment and remaining useful life prediction of aero-engine based on BiGRU and MMoE. *Reliab Eng Syst Saf* 2022;220:108263. <http://dx.doi.org/10.1016/j.res.2021.108263>.
- [24] Li X, Zhang W, Ma H, Luo Z, Li X. Data alignments in machinery remaining useful life prediction using deep adversarial neural networks. *Knowl-Based Syst* 2020;197:105843. <http://dx.doi.org/10.1016/j.knsys.2020.105843>.
- [25] Xiao L, Tang J, Zhang X, Bechhoefer E, Ding S. Remaining useful life prediction based on intentional noise injection and feature reconstruction. *Reliab Eng Syst Saf* 2021;215:107871. <http://dx.doi.org/10.1016/j.res.2021.107871>.
- [26] Mo Y, Wu Q, Li X, Huang B. Remaining useful life estimation via transformer encoder enhanced by a gated convolutional unit. *J Intell Manuf* 2021;32(2). <http://dx.doi.org/10.1007/s10845-021-01750-x>.
- [27] Dong S, Xiao J, Hu X, Fang N, Liu L, Yao J. Deep transfer learning based on Bi-LSTM and attention for remaining useful life prediction of rolling bearing. *Reliab Eng Syst Saf* 2023;230:108914. <http://dx.doi.org/10.1016/j.res.2022.108914>.
- [28] Vaswani A, Shazeer N, Parmar N, Uszkoreit J, Jones L, Gomez AN, et al. Attention is all you need. In: Advances in neural information processing systems. 2017, p. 5998–6008. <http://dx.doi.org/10.48550/arXiv.1706.03762>.
- [29] Zhou H, Zhang S, Peng J, Zhang S, Li J, Xiong H, et al. Informer: Beyond efficient Transformer for long sequence time-series forecasting. In: Proceedings of AAAI. 2021, <https://ojs.aaai.org/index.php/AAAI/article/view/17325>.
- [30] Chen X, Zhang Z, Zhang Z. Real-time equipment condition assessment for a class-imbalanced dataset based on heterogeneous ensemble learning. *Eksploracja I Niezawodność* 2019;21(1). <http://dx.doi.org/10.17531/ein.2019.1.9>.
- [31] Yan J, He Z, He S. Multitask learning of health state assessment and remaining useful life prediction for sensor-equipped machines. *Reliab Eng Syst Saf* 2023;109141. <http://dx.doi.org/10.1016/j.res.2023.109141>.
- [32] Zhang W, Yang D, Xu Y, Huang X, Gidlund M. DeepHealth: A self-attention based method for instant intelligent predictive maintenance in industrial internet of things. *IEEE Trans Ind Inf* 2021;17(8):5461–73. <http://dx.doi.org/10.1109/TII.2020.3029551>.
- [33] Lin T-Y, Goyal P, Girshick R, He K, Dollár P. Focal loss for dense object detection. In: Proceedings of the IEEE international conference on computer vision. 2017, p. 2980–8. <http://dx.doi.org/10.48550/arXiv.1708.02002>.
- [34] Kingma DP, Ba J. Adam: A method for stochastic optimization. 2014, <http://dx.doi.org/10.48550/arXiv.1412.6980>, arXiv preprint [arXiv:1412.6980](http://arxiv.org/abs/1412.6980).
- [35] Cao Y, Jia M, Ding P, Ding Y. Transfer learning for remaining useful life prediction of multi-conditions bearings based on bidirectional-GRU network. *Measurement* 2021;178(5):109287. <http://dx.doi.org/10.1016/j.measurement.2021.109287>.